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MAGNETOSTRATIGRAPHY OF THE IMNAHA AND GRANDE RONDE BASALTS IN THE SOUTHEAST PART OF THE COLUMBIA PLATEAU

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ABSTRACT. Two hundred and thirty-one oriented samples were collected from basalt flows at Dug Bar (OR), Fall Creek (OR), Grande Ronde (WA), Joseph Creek (WA), and Rocky Canyon (ID), the stratigraphy of which has been established by a detailed field and petrochemical study. The stable remanent magnetism of the samples, when compared to the magnetic polarities determined independently using a portable fluxgate magnetometer, confirm the fluxgate magnetometer polarity results in all cases but clarify the presence of flows whose polarity is transitional, with low latitude virtual magnetic poles.

The results demonstrate that the Imnaha Basalt belongs to a single normal polarity epoch (N_0), except for a few basal flows with reversed polarity (R_0) and the top few flows, which have a transitional polarity. The abrupt chemical change from Imnaha to Grande Ronde Basalt occurred during the N_0/R_1 polarity transition, with the earliest two or three flows of Grande Ronde Basalt possessing transitional polarity and the subsequent flows reversed (R_1), normal (N_1), and then reversed (R_2) polarities.

The five polarity epochs are sequential. In the center of the Columbia Plateau eruption appears to have continued through the next two polarity epochs (N_2 and R_3). These seven polarity epochs, represented by the lower part of the Columbia River Basalt Group (Imnaha, Grande Ronde, and Wanapum Basalts) and accounting for over 99 percent by volume of the eruption, are tentatively correlated with magnetic anomalies in epochs 14, 15, and 16 of deep-sea sedimentary cores.

Field and geochemical evidence implies that the normal to reversed magnetic polarity sequence of the Picture Gorge Basalt cannot be correlated with the N_0/R_1 sequence of the Imnaha/Grande Ronde Basalt but rather with the N_1/R_2 polarity sequence in the middle of the Grande Ronde Basalt eruption.

The mean paleomagnetic poles for the areas studied are not significantly different from the middle Miocene poles from stable areas of the continent and permit only minor tectonic rotation since that time.

INTRODUCTION

It has been demonstrated in several volcanic regions that the remanent magnetism of basalt flows can provide distinctive stratigraphic indices (for example, Watkins and Walker, 1977). Remanent magnetization directions can be obtained at two levels of precision: by use of laboratory measurements, which give directions within a few degrees, or by use of field polarity measurements only. The former are obtained through laboratory analyses of 4 to 8 oriented specimens taken from each flow. While the results can be definitive in showing that given flows are *not* time correlative (should different directions be recorded), they do not necessarily imply definite time equivalents if similar directions

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are observed. This is because the geomagnetic field can possess similar direction at different times, reflecting the fact that the geomagnetic pole occupies high latitudes for at least 90 percent of the time. Should, however, a geomagnetic field excursion or polarity transition (in which the virtual geomagnetic pole is in tropical latitudes) be recorded, then demonstration of true time equivalence is potentially much more probable, because of the much smaller time the geomagnetic field spends in such a configuration. A polarity transition is by definition the period between opposite polarities and lasts between 1000 and 5000 yrs (Opdyke, 1972), whereas an excursion refers to a possibly shorter period during which a lower latitude pole occurs without any change in polarity. The period between successive polarity changes during the late Cenozoic has averaged between 140,000 and 190,000 yrs (Watkins and Walker, 1977).

Measurement of polarity is a more simple exercise, which can be done in the field with a portable fluxgate magnetometer (Doell and Cox, 1962). The resulting polarity stratigraphy has limitations, because one normal polarity epoch cannot be distinguished from another without additional control, although no difficulties exist in proving *non*-correlations. An additional problem with field measurement of polarity is that an original reversed polarity can be overprinted by unstable components, which reflect the normal polarity of the present-day geomagnetic fields. Lavas with such "pseudo-normal" polarity have been recorded on the Columbia Plateau (Watkins, 1965), and their true reversed polarity recognized following removal of unstable components by standard laboratory techniques. The magnetic stability of basalts depends on the oxidation state of the titanomagnetites (Watkins and Haggerty, 1967). Low oxidation states yield low stabilities, whereas the higher oxidation states far more readily retain the original magnetic signature, so that samples from the more oxidized lower and upper margins of flows are more likely to provide stable signals (Watkins and Baksi, 1974; Swanson and Wright, 1976a).

Currently the stratigraphy of the Columbia River Basalt Group (fig. 1) is being studied intensively. Various parties are cooperating in detailed field mapping, petrochemical analyses, and magnetic polarity measurements (Nathan and Fruchter, 1974; Camp, ms; Hooper, Kleck, and Knowles, 1976; Swanson and Wright, 1976a, b; Vallier and Hooper, 1976; Swanson and others, 1977). The result is a detailed stratigraphic column of the basalt (fig. 2; Swanson and others, in press) which can be correlated across the Columbia Plateau. In this paper we present the results of a paleomagnetic survey and corresponding laboratory analyses of five stratigraphically critical sections through the basalt in the southeast part of the Columbia Plateau. These provide a test of the paleomagnetic zones mapped in the field using a much larger number of fluxgate magnetometer readings and allow a proper comparison with the earlier determinations of the stable remanent magnetism from the Plateau by Watkins and Baksi (1974). The results are also relevant to regional tectonic problems.

GEOLOGICAL BACKGROUND

Eighteen hundred meters of the lower part of the Columbia River Basalt Group are well exposed in the deep canyons of the Snake and Salmon Rivers in the tri-state area of Washington, Oregon, and Idaho (fig. 1). The composite section includes over 20 flows of Imnaha Basalt overlain by a minimum of 50 flows of the Yakima Basalt Subgroup including the Grande Ronde Basalt (lower Yakima basalt), the Wanapum Basalt (middle Yakima basalt), and the Saddle Mountains Basalt (upper Yakima basalt) (fig. 2). The age of the Grande Ronde and Wanapum Basalts is between 13.7 and 16.0 m.y. (Watkins and Baksi, 1974), whereas Saddle Mountains Basalt spans a much greater time range of about 14 to 6 m.y. (McKee, Swanson, and Wright, 1977).

Mapping along the well exposed canyon walls, combined with petrochemical studies of measured sections, has established the flow stratigraphy to the extent that most flows can now be recognized either individually or as a member of a small coherent group of two or three. The flows have been traced across an area bounded by Lewiston (ID) in the east (Reidel, 1975 and ms; Camp, ms; Kleck, ms; Holden and Hooper, 1976). The area extends 120 km in a north-northwest direction along the Grande Ronde feeder dike swarm and 60 km east of the dike swarm (Taubeneck, 1970; Price, 1974).

Polarity measurements using a field fluxgate magnetometer were incorporated into field mapping and flow correlation, particularly in the Grande Ronde Basalt, where chemical and mineralogical differences between flows are small (Camp, ms; Hooper, Kleck, and Knowles, 1976). Field polarity measurements are based on a minimum of three samples at each outcrop. Whereas occasional flows in some parts of the section give conflicting polarities, a coherent pattern of magnetic polarity has emerged for the Imnaha Basalt and Yakima Basalt Subgroup (fig. 2; Nathan and Fruchter, 1974; Camp, 1976; Hooper, Kleck, and Knowles, 1976; Swanson and Wright, 1976a, b; Vallier and Hooper, 1976; Swanson and others, 1977 and in press). Geologists working on the Columbia Plateau have agreed informally that the reversed polarity epoch represented by the oldest flows of Grande Ronde Basalt in the type section should be labelled R_1 and that subsequent epochs in the Grande Ronde Basalt be labelled N_1, R_2, and N_2 (Swanson and others, 1977 and in press).

In its lower part this field-based but consistent magnetostratigraphic pattern conflicted with the earlier more detailed magnetic studies of Watkins and Baksi (1974). These authors concluded (1974, fig. 16) that the upper members of the Imnaha Basalt had reversed polarity while the lowest members of the Grande Ronde Basalt in this southeast part of the Plateau had normal polarity. To resolve this conflict five well studied basalt sections were chosen for more detailed magnetic investigation. The type section for the Imnaha Basalt at Dug Bar (DB), sections across the Imnaha-Grande Ronde Basalt contact at Fall Creek (FL) and Joseph Creek (GRJ), and the type section for the Grande Ronde Basalt

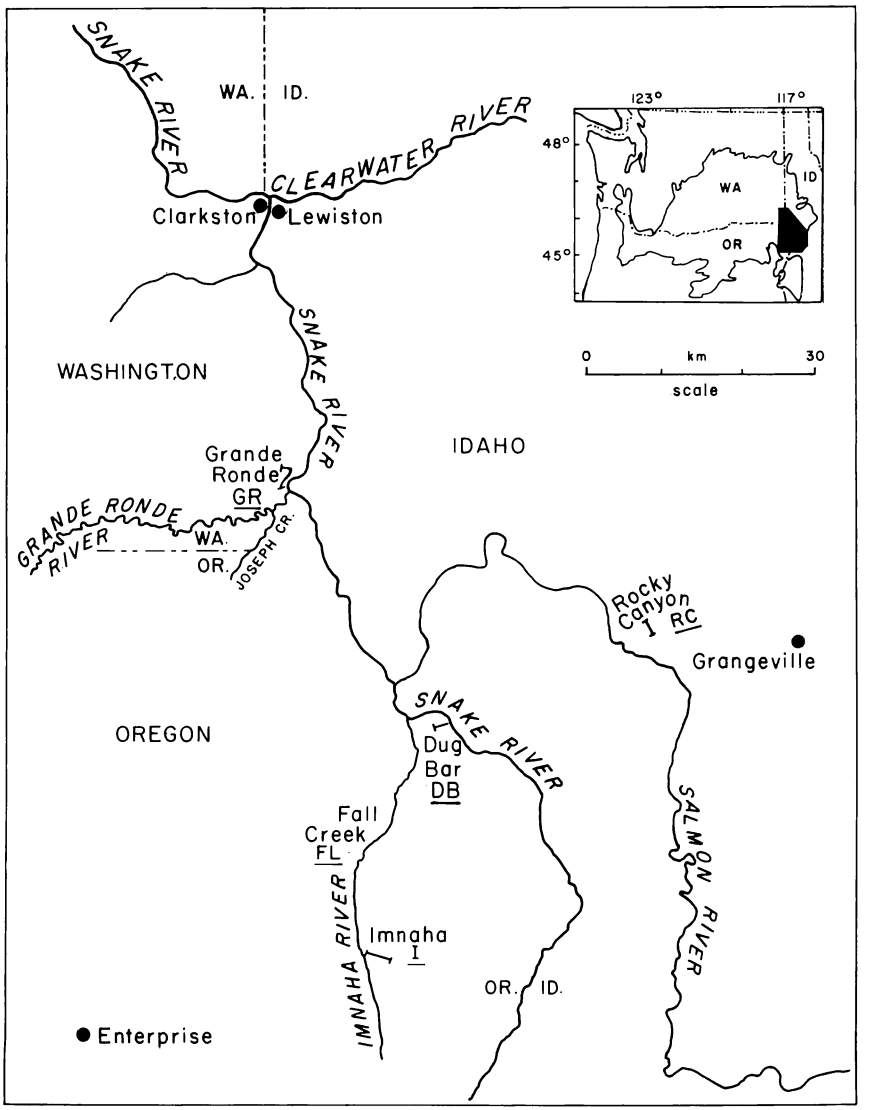


Fig. 1. Sketch map showing location of sampled sections.

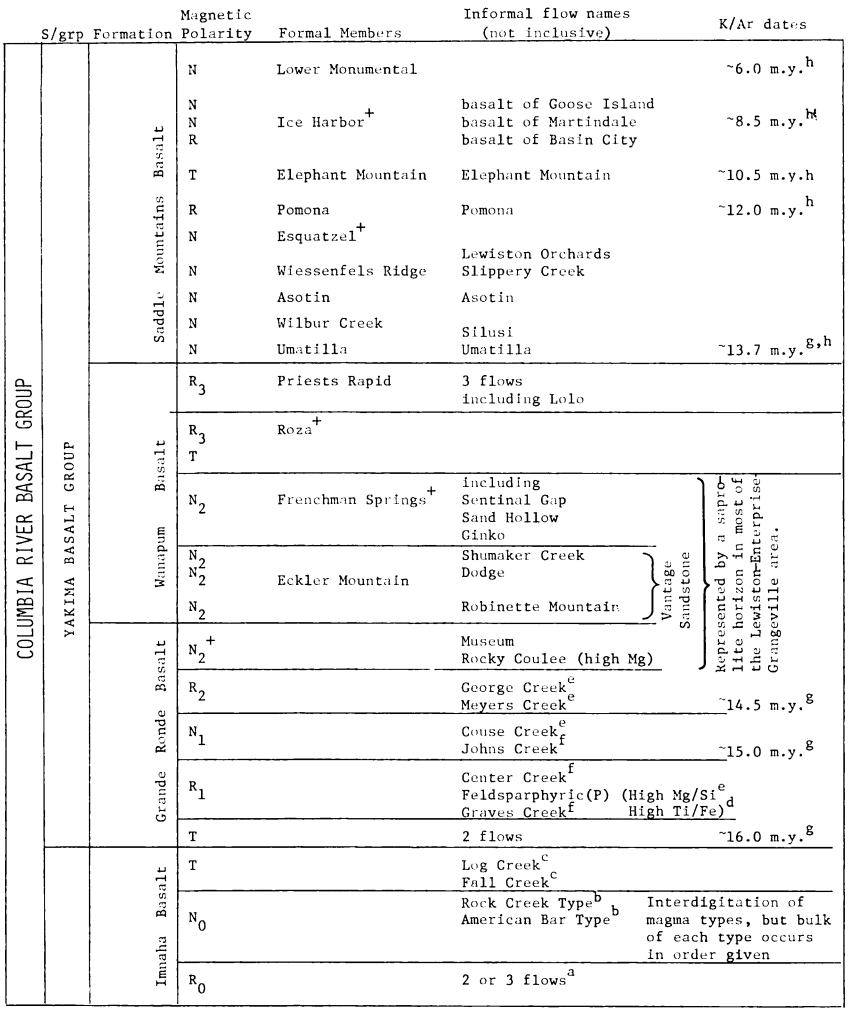


Fig. 2. The stratigraphy of the Columbia River Basalt Group, modified from Swanson and others (in press) with informal flow names used within the area studied. Additions from [a]W. H. Taubeneck, D. A. Swanson and D. O. Nelson, personal communication, 1977. [b]Hooper, Kleck, and Knowles, 1976; [c]Kleck, 1976; [d]Reidel, 1978; [e]Camp, 1976; [f]Bond 1963; [g]Watkins and Baksi, 1974; [h]McKee and others, 1977. The Picture Gorge Basalt, in the John Day Basin of north central Oregon, is interbedded with Grande Ronde flows of the N_1 and R_2 polarity epochs (D. A. Swanson and R. D. Beneley, oral comm., 1977).

+ Absent in Clarkston-Enterprise-Grangeville area.

at the mouth of the Grande Ronde River (GR) lie within the Grande Ronde dike swarm (fig. 1); the Rocky Canyon (RC) section was chosen because it covers the same stratigraphic interval (Holden, ms; Holden and Hooper, 1976) but is 60 km east of the main dike swarm and is fed at least in part by its own local dike system (Price, Knowles, and Bond, 1973).

METHODS AND RESULTS

A total of 231 cores of 2.5 cm diameter and 8 cm average length were drilled using a gasoline-powered portable drill. This represents an average of between 5 and 6 separate cores from each flow. The cores were oriented while attached to outcrop, using a sun compass in most instances. The number of flows in each sequence is given in table 1.

A specimen of 2.2 cm length was sliced from each core for laboratory analysis. The direction and intensity of natural remanent magnetism (NRM) of each specimen was measured using a Digico spinner magnetometer. The mean results for each flow are given in table 1A. Unstable components were minimized by treatment in peak alternating magnetic fields of 100 oersteds, which was followed by remeasurement of the direction and intensity of remanent magnetism. The procedure was repeated using a peak alternating field of 200 oersteds. The values used to compute the mean direction for each flow were selected on a minimum scatter basis (Watkins and Richardson, 1968), obtained in virtually all cases after the higher field treatment. The results are given in table 1B and figures 3, 4, and 5.

DISCUSSION

Magnetic Properties

Three flows in the Rocky Canyon section have pseudo-normal polarity (RC 9, 11, and 16) in the sense that their NRMs give normal polarities before demagnetization but reversed after demagnetization with virtual geomagnetic poles (VGP) at high latitudes (table 1). All three flows were correctly recorded as having reversed polarity on the fluxgate magnetometer, probably because of the tendency to collect samples for the fluxgate magnetometer from the more friable and also more oxidized lower and upper margins of a flow (Watkins and Baksi, 1974; Swanson and Wright, 1976a), while clean cores for the more detailed laboratory study are most easily recovered from the denser flow interiors (Watkins and Haggerty, 1967).

Two flows at Fall Creek (FL 2, 3), one near the base of the Grande Ronde section (GR 11), and six of the seven flows at Joseph Creek (GRJ 1 to 7) have virtual geomagnetic poles at low latitudes (figs. 3, 4, and 5) and are regarded as having transitional polarity. The remainder of the flows examined provide coherent directions of remanent magnetism with virtual geomagnetic poles at high latitudes.

TABLE 1

A. Mean directions of natural remanent magnetism for each lava

(I) GR (Grande Ronde)

NO	N	D	I	R	θ^1	ϕ^1
GR 10	6	226.0	-40.6	5.8922	-46.6	168.9
GR 12	6	166.7	-61.8	5.9533	-80.0	319.6
GR 13	7	156.0	-79.3	6.9810	-63.9	44.0
GR 14	7	190.1	-62.6	6.9933	-82.5	166.5
GR 15	6	183.9	-70.9	5.9883	-80.5	76.6
GR 16	6	150.1	-70.8	5.8011	-69.2	9.8
GR 17	5	169.6	-61.1	4.9921	-81.5	309.0
GR 18	4	15.7	44.2	3.9550	66.2	25.8
GR 19	6	308.8	48.8	5.9545	47.3	149.0

(II) RC (Rocky Canyon)

NO	N	D	I	R	θ^1	ϕ^1
RC 1	6	351.9	48.7	5.9813	72.5	87.8
RC 2	2	340.8	57.1	1.9937	73.6	130.5
RC 3	6	342.5	67.5	5.9835	77.6	180.7
RC 4	6	335.3	68.5	5.9722	72.8	182.4
RC 5	6	355.8	59.5	5.5128	83.6	93.7
RC 6	4	306.5	62.6	3.9693	52.8	170.5
RC 7	6	168.8	-74.9	5.9923	-73.0	45.2
RC 9	7	193.4	-55.2	6.9877	-75.6	194.1
RC 10	5	169.5	-73.1	4.9975	-75.8	40.9
RC 11	5	164.2	-65.0	4.9794	-79.1	345.0
RC 12	6	160.8	-55.1	5.9471	-72.3	304.9
RC 13	6	3.8	60.2	5.9855	84.5	32.5
RC 14	6	325.1	56.7	5.9530	62.8	147.6
RC 15	5	31.2	40.6	4.8759	56.1	4.9
RC 16	6	169.1	-61.8	5.9830	-82.0	315.9
RC 17	7	10.1	84.3	6.7006	57.0	247.2

(III) DB (Dug Bar)

NO	N	D	I	R	θ^1	ϕ^1
DB 2	5	349.2	55.1	4.9968	77.0	105.8
DB 3	7	13.2	22.9	6.9326	54.3	40.7
DB 4	4	330.4	60.8	3.9534	68.4	153.4
DB 5	6	338.2	61.4	5.9824	74.1	149.4
DB 6	2	21.2	62.0	1.9989	74.8	335.3
DB 7	7	26.8	59.6	6.9546	69.8	339.2
DB 8	6	355.6	61.0	5.9728	85.1	105.4

(IV) FL (Fall Creek)

NO	N	D	I	R	θ^1	ϕ^1
FL 1	6	199.2	-67.4	5.9834	-76.5	127.7
FL 2	4	74.3	-76.2	3.9188	-34.1	32.8
FL 3	5	331.1	-45.7	4.7987	12.5	89.8

(V) GRJ (Joseph Creek)

NO	N	D	I	R	θ^1	ϕ^1
GRJ 1	6	30.8	-66.9	5.0572	-9.2	277.5
GRJ 2	5	32.7	-57.0	4.9619	1.4	271.7
GRJ 3	6	61.5	-72.2	5.9672	-25.2	265.4
GRJ 4	2	169.6	-77.4	1.9999	-69.3	285.0
GRJ 5	6	71.3	16.0	5.6844	18.8	215.0
GRJ 6	6	260.2	-62.7	5.8626	-35.8	358.0
GRJ 7	6	69.8	9.2	5.9933	17.2	218.6

B. Mean remanent magnetism results for each lava after demagnetization treatment

(I) GR (Grande Ronde)

NO	N	D	I	R	θ^1	ϕ^1
GR 10	6	236.3	-33.8	5.3861	-36.4	164.5
GR 12	6	166.8	-62.7	5.9555	-80.5	325.9
GR 13	7	122.8	-79.4	6.9833	-53.7	33.3
GR 14	7	175.2	-56.4	5.8738	-80.2	266.1
GR 15	6	197.7	-72.7	5.7982	-73.9	98.6
GR 16	6	170.0	-51.4	4.9443	-74.0	275.2
GR 17	5	162.6	-60.3	4.9769	-76.6	318.3
GR 18	4	8.6	46.9	3.9798	70.8	39.6
GR 19	6	341.4	58.7	5.9861	74.9	133.8

(II) RC (Rocky Canyon)

NO	N	D	I	R	θ^1	ϕ^1
RC 1	6	2.5	58.6	5.8576	83.2	47.2
RC 2	2	343.2	61.8	1.9910	77.7	145.9
RC 3	6	328.0	61.7	5.7531	67.1	157.6
RC 4	6	350.7	68.3	5.9832	81.7	199.5
RC 5	6	22.1	73.5	5.4663	71.2	280.1
RC 6	4	329.3	74.3	3.9488	66.8	204.1
RC 7	6	161.4	-77.4	5.9628	-67.6	43.6
RC 9	7	296.7	75.0	6.7511	51.3	201.0
RC 10	5	165.8	-72.7	4.9902	-75.0	33.6
RC 11	6	34.6	47.3	4.2213	57.7	854.3
RC 12	6	157.2	-52.3	5.9379	-68.2	305.0
RC 13	6	5.0	60.0	5.9928	83.8	26.3
RC 14	6	347.7	63.0	5.9121	81.2	148.9
RC 15	5	19.3	53.4	4.9811	71.1	5.9
RC 16	6	346.7	-2.8	5.9554	41.3	81.4
RC 17	7	18.1	68.3	6.8850	76.9	302.1

(III) DB (Dug Bar)

NO	N	D	I	R	θ^1	ϕ^1
DB 2	5	352.6	60.5	4.9714	83.1	116.7
DB 3	7	0.6	48.0	6.8569	73.3	61.6
DB 4	4	227.9	83.3	3.5883	36.3	231.1
DB 5	6	344.3	59.3	5.9893	77.2	132.7
DB 6	2	15.2	64.4	1.9965	79.5	325.2
DB 7	7	5.0	57.4	6.8433	81.4	35.8
DB 8	6	339.3	68.9	5.6648	75.0	186.6

(IV) FL (Fall Creek)

NO	N	D	I	R	θ^1	ϕ^1
FL 1	6	202.1	-69.8	5.9716	-73.8	116.7
FL 2	4	350.7	-58.6	3.1589	4.4	70.8
FL 3	5	335.3	-21.5	4.9247	28.9	91.6

(V) GRJ (Joseph Creek)

NO	N	D	I	R	θ^1	ϕ^1
GRJ 1	6	9.5	-57.5	5.4336	5.5	269.6
GRJ 2	5	330.6	48.0	1.7605	61.5	0.9
GRJ 3	6	131.4	47.7	3.5781	-3.3	158.3
GRJ 4	2	149.8	-66.6	1.9999	-69.5	226.9
GRJ 5	6	55.8	41.4	5.7991	40.3	213.9
GRJ 6	6	25.2	48.0	4.7466	64.0	239.1
GRJ 7	6	64.8	18.9	5.9812	24.4	218.8

No. = flow number. (See fig. 3 for stratigraphic details). N = number of separate specimens taken from given lava. D and I = declination and inclination of mean remanent magnetism in degrees east of north and with respect to horizontal (— = above horizontal, no sign = below horizontal), respectively. R = resultant vector, applying unit vector per specimen (most conventional parameters can be derived using N and R). θ and ϕ = latitude and longitude of virtual geomagnetic pole, in degrees north or south (— = south), and east of Greenwich, respectively.

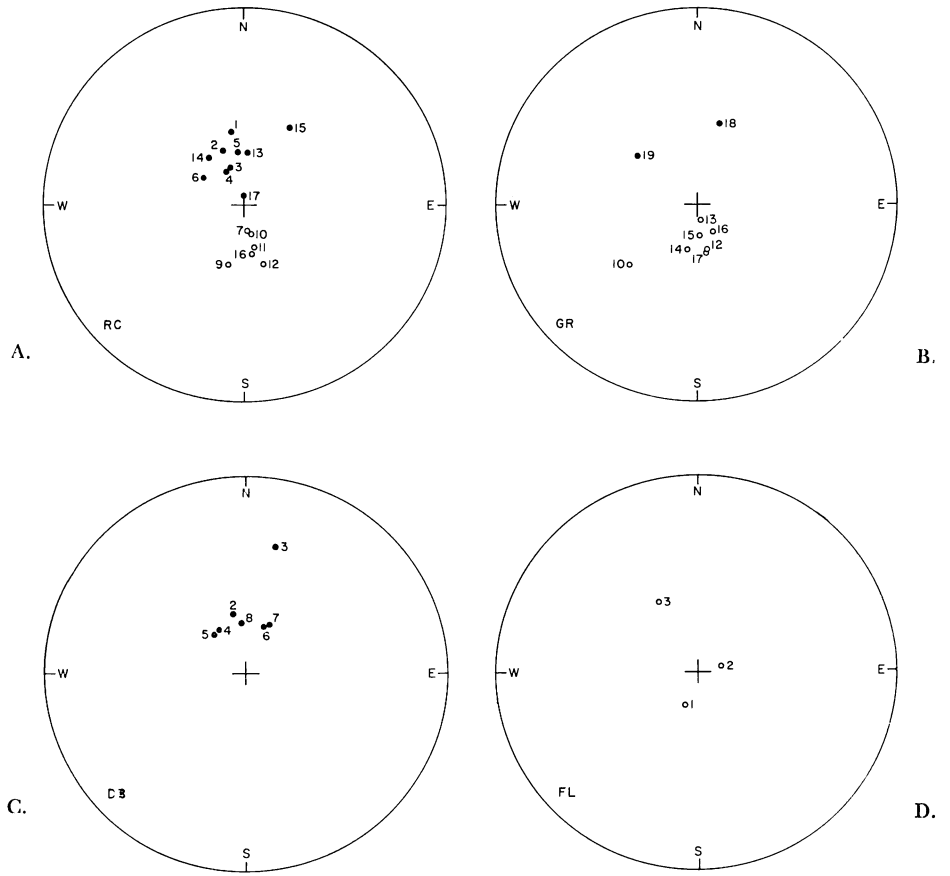


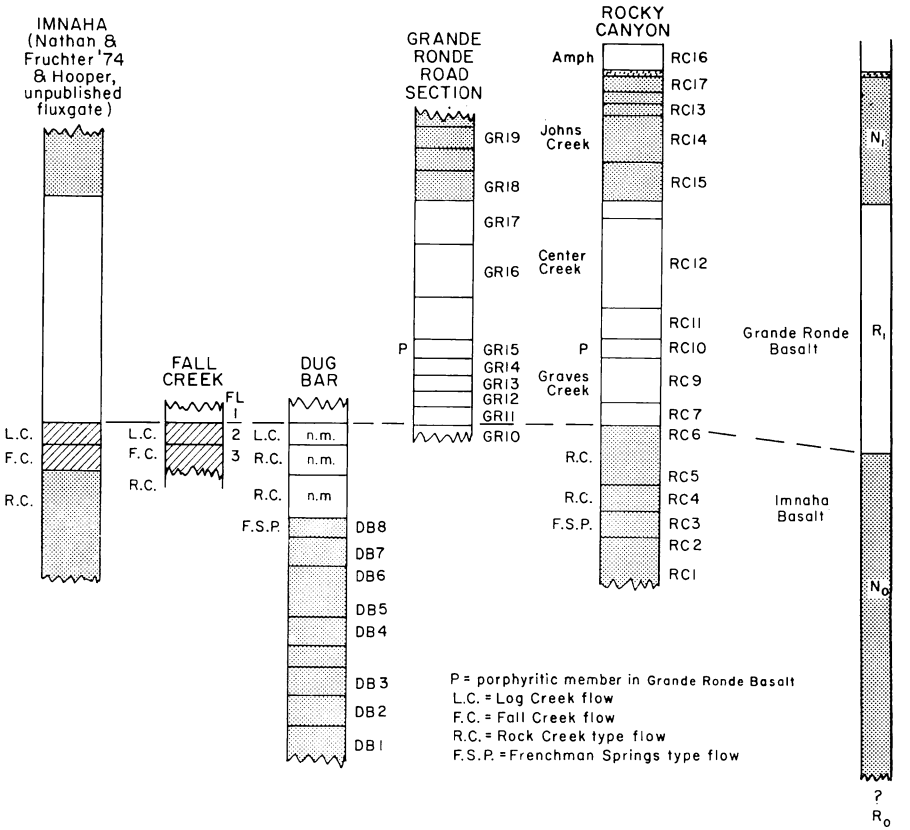
Fig. 3. Stereographic projections of mean directions of remanent magnetism after demagnetization for each lava flow. (A) Rocky Canyon Section: numbers 1 to 6 are Imnaha Basalt flows (N_0 epoch); 7 and 9 to 12 are Grande Ronde Basalt (R_1 epoch); 13, 14, 15, 17 are Grande Ronde Basalt (N_1); 16 is a younger flow, unconformable above the rest of the section. (B) Grande Ronde Section: number 10 = youngest flow of Imnaha Basalt; 12 to 19 are successive flows above the Imnaha Basalt/Grande Ronde Basalt contact. (C) Dug Bar Section: numbers 2 to 8 are N_0 epoch. (D) Fall Creek Section: numbers 2 and 3 are Imnaha Basalt, with anomalous directions yielding low latitude virtual geomagnetic poles; number 1 is Grande Ronde Basalt (R_1 epoch).

In all diagrams solid circle = lower hemisphere; open circle = upper hemisphere. For original data, see table 1B.

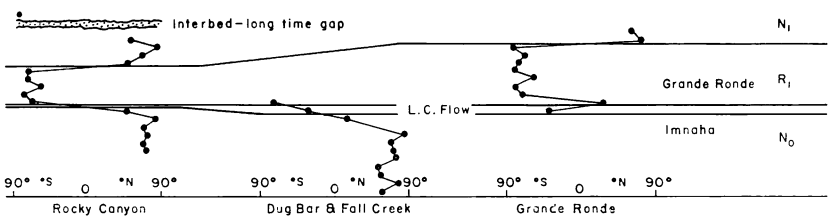
Magnetostratigraphy

The polarity of the flows in the main sections is shown in figure 4A, and the VGP latitudes are plotted in figure 4B. The stratigraphic correlation is that determined using field, petrochemical, and portable fluxgate magnetometer measurements. The polarity indicated is that determined in the laboratory examination of the cores following demagnetization (table 1B), with the exception of the Imnaha section in which the polarity is that determined by Nathan and Fruchter (1974) and ourselves using a fluxgate magnetometer. The earlier data of Watkins and Baksi

Fig. 4



A. Magnetostratigraphy of the Columbia River Basalt in the Snake and Salmon River Canyons. Magnetostratigraphy of the four sections, Dug Bar (DB), Fall Creek (FL), Grande Ronde (GR), and Rocky Canyon (RC) was measured on the outcrops using a fluxgate magnetometer and in the laboratory using oriented samples (flow numbers correspond to those in table 1 and fig. 3). The Imnaha section was measured using a fluxgate magnetometer only (Nathan and Fruchter, 1974; and present study). Stiple represents normal polarity, white represents reversed polarity, oblique lines represent transitional polarity, and short dashes represent a sedimentary interbed. n. m. = not measured.



B. Variation of latitude of virtual geomagnetic pole with stratigraphic position for Rocky Canyon, Dug Bar, and Fall Creek and Grande Ronde sections. See table 1 for original data. LC = Log Creek Member.

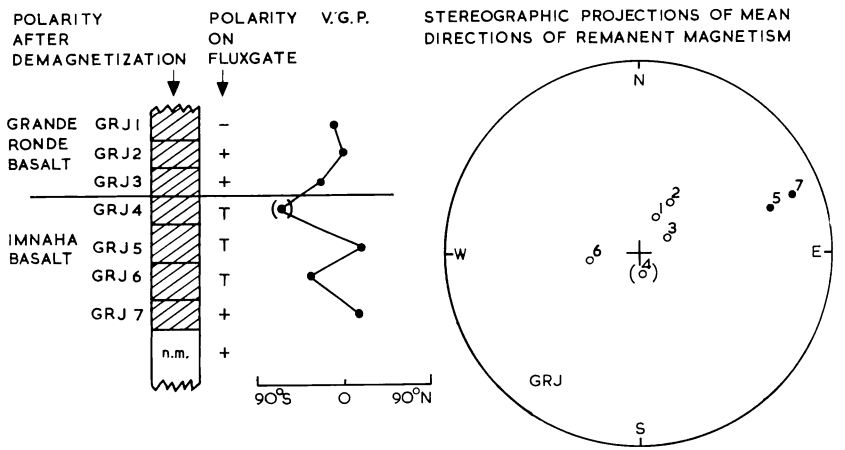


Fig. 5. Mean remanent magnetic poles after demagnetization of seven flows spanning the Imnaha-Grande Ronde Basalt contact in Joseph Creek. Symbols as in figures 3 and 4. GRJ 4 is placed in parentheses, because only two cores from an altered and zeolitized scoria were recovered.

(1974, figs. 6, 7, 8, 9, and 16) indicate a less simple magnetostratigraphy, but their interpretation is now known to be incorrect owing to slumping and faulting. Many other stratigraphic sections through this part of the basalt succession confirm the same simple polarity pattern apparent in figure 4A.

The Imnaha Basalt.—All flows of the Imnaha Basalt exposed in the area have normal polarity except for the top few flows where the Grande Ronde dike swarm is most dense. These few upper Imnaha Basalt flows have directions of remanent magnetism, which do not give VGPs close to the expected pole positions, and lie between flows of the N_0 polarity epoch and those of the R_1 polarity epoch, consistent with the recording of a polarity transition.

West of the mapped area, in the Wallowa Mountains, W. H. Taubeneck, D. A. Swanson, and D. O. Nelson (personal commun., 1977) report a thick sequence of Imnaha Basalt flows for which fluxgate magnetometer readings indicate normal polarity, with the exception of two or three reversed flows at the base (R_0) and several thin reversed flows at the top.

The Grande Ronde Basalt.—In most stratigraphic sections the lowest member of the Grande Ronde Basalt has a reversed polarity. An exception is the lowest Grande Ronde Basalt flow, resting on Imnaha Basalt, in the Grande Ronde section (GR 11, fig. 4B). In the nearby Joseph Creek (WA) section (fig. 1) field fluxgate measurements suggest the top three flows of Imnaha Basalt are transitional and the bottom two Grande Ronde Basalt flows have normal polarity, higher flows being clearly reversed. But demagnetization of cored samples (GRJ, fig. 5) indicates that a sequence of seven flows possesses the low latitude VGP positions consistent with a polarity transition.

The Joseph Creek section shows that the abrupt change in magma composition from the Imnaha to the Grande Ronde Basalt occurred *during* the polarity transition from N_0 to R_1. This section occurs in an area where the frequency of feeder dikes is higher than in any other part of the region being discussed and contains at least four flows not present in any other measured section. Thus, in the densest part of the Grande Ronde dike swarm many additional flows which do not extend any significant distance to the east are present. At Rocky Canyon, 60 km east of the main dike swarm, where the flows are fewer and thicker, the transition from N_0 to R_1 is sharp, with no evidence of transitional magnetic orientations (fig. 4B).

The magnetic reversal from R_1 to N_1 is sharp and occurs between two flows within the Grande Ronde Basalt in both the Grande Ronde and the Rocky Canyon sections. No flow with transitional magnetic orientations was observed. More flows are present in the R_1 epoch in those areas in which feeder dikes are most abundant.

In general, the demagnetization of the cored samples has confirmed the field fluxgate measurements and established the occurrence of flows with transitional polarity. Whereas fluxgate readings in the main part of the Imnaha Basalt are strongly normal, those in the R_1 polarity epoch may give give weak and occasionally conflicting results for an individual flow, almost certainly reflecting normal polarity overprinting in parts of the flows.

Comments on Regional Magnetostratigraphy

Reidel (1975 and ms), Camp (ms), and Holden and Hooper (1976) have shown that the mineralogy and chemical composition of the flows at the base of the R_1 polarity epoch can be distinguished from those of the Grande Ronde basalt flows falling within the upper part of the R_1, the N_1, and the R_2 polarity epochs, so that using all techniques currently available it should be possible to distinguish between basalts of the R_1 and R_2 epochs in other parts of the Columbia Plateau, even where the succession is complicated by faulting and poor exposure, as in the Wallowa and Blue Mountains.

The N_1/R_2 magnetic reversal (fig. 2) has not been studied by detailed coring, but it has been mapped throughout the Lewiston Basin by Camp (ms) and to the west by Swanson and others (1977). Flows belonging to the N_2 polarity epoch of the Grande Ronde Basalt are not exposed in the area studied, and the Frenchman Springs and Roza Members from the lower part of the Wanapum Basalt are also missing, the period of their extrusion being represented by a major saprolite horizon (fig. 2; Camp, ms). The Lolo flow (Priests Rapid Member) at the top of the Wanapum Basalt formation and the various members of the Saddle Mountains Basalt exposed in the Lewiston Basin and to the north (Camp, ms; Swanson and others, 1977) are all separated by thin sedimentary interbeds. This and the much larger gaps between the ages of the dated flows (McKee, Swanson, and Wright, 1977) imply that

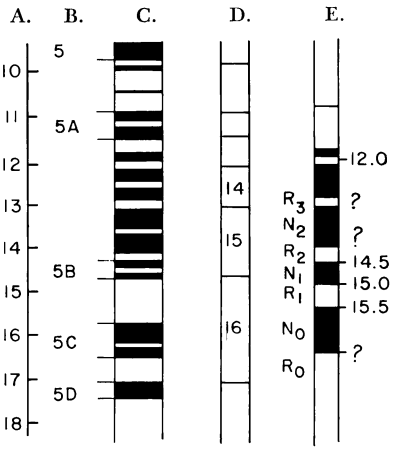


Fig. 6. Postulated correlation between epochs R_0 and R_3 and the polarity time scale. (A) age in m.y.; (B) marine magnetic anomaly number; (C) polarity history (black = normal, clear = reversed); (D) polarity epoch number; (E) polarity epochs and radiometric ages for the lower part of the Columbia River Group (from present study and Watkins and Baksi, 1974). (A) through (D) are from LaBreque, Kent, and Cande (1977).

eruptions were becoming less frequent with time, and it is certain that a number of polarity epochs are missing from this part of the stratigraphic column.

Over much of the Columbia Plateau to the west of the area described in this report a small number of magnesium-rich flows of the N_2 polarity epoch form the top of the Grande Ronde Basalt sequence (Taylor, ms; Swanson and others, in press). These are separated from the overlying Frenchman Springs Member (normal polarity) by either an interbed or a saprolite (fig. 2; Swanson and others, in press). But some 40 km west of Clarkston, Swanson and others (1977) have mapped flows between the Grande Ronde Basalt and Frenchman Springs Member which they call the Eckler Mountain Basalt. At one locality the three types, Grande Ronde, Eckler Mountain, and Frenchman Springs, are interbedded (Swanson and others, 1977 and in press). All these flows have normal polarity, and so it is probable that the Frenchman Springs Member is part of the N_2 polarity epoch (Choiniere and Swanson, 1979).

The transitional and reversed polarity of the Roza Member (Choiniere and Swanson, 1979) and reversed polarity of the Priests Rapid Member also appear to follow consecutively, and we suggest, as do Choiniere and Swanson, that the latter be assigned to the informal R_3 polarity epoch. Thus at least five (N_0, R_1, N_1, R_2, N_2) and very probably seven consecutive polarity epochs (R_0, N_0, R_1, N_1, R_2, N_2, R_3) are represented by the lower part of the Columbia River Basalt Group.

Earlier workers have assumed that the Picture Gorge Basalt and the Imnaha Basalt were broadly contemporaneous because of the apparent similarity in mineral and major element composition (Waters,

1961). The normal polarity of the larger part of the Picture Gorge Basalt sequence, changing near the top to reversed polarity, was correlated by both Nathan and Fruchter (1974) and Watkins and Baksi (1974) to the N_0-R_1 polarity epochs of this paper. It is now known that the Picture Gorge Basalt and the Imnaha Basalt have significantly different major and trace element compositions (Wright, Grolier, and Swanson, 1973; Hooper, 1974; Nathan and Fruchter, 1974; Hooper, Kleck, and Knowles, 1976; McDougall, 1976). K–Ar dates (Watkins and Baksi, 1974) demonstrate that the range of Picture Gorge Basalt dates (15.2 ± 0.6 m.y.) fall within the range (14.0-16.0 m.y.) for the Grande Ronde Basalt. Further, we know of no flows with good reversed polarity at the top of the Imnaha Basalt sequence, although transitional polarities are present, and field fluxgate measurements of reversed flows are reported from farther west by W. H. Taubeneck, D. A. Swanson, and D. O. Nelson (personal commun., 1977). Even more critical to the relative age of the Picture Gorge Basalt is the demonstration by Cockerham and Bentley (1973) and Nathan and Fruchter (1974) that flows of Picture Gorge Basalt are interbedded with those of Grande Ronde Basalt in the Butte Creek area of the Blue Mountains in north central Oregon. Recent mapping and fluxgate work by D. A. Swanson and R. D. Bentley (oral commun., 1977) in the same area has established that the flows of Grande Ronde Basalt with which those of the Picture Gorge are interbedded belong to the N_1 and R_2 polarity epochs. Thus the present evidence strongly suggests that most or all the Picture Gorge Basalt is younger than the Imnaha Basalt. K–Ar dates for the Imnaha Basalt are not yet available.

Correlation With the Polarity Time Scale

Available K/Ar dates of the Grande Ronde Basalt have low precision relative to the short periods of time occupied by the magnetic epochs (Gray and Kittleman, 1967; Holmgren, 1970; Watkins and Baksi, 1974). The dates most relevant to this discussion and those most consistent with the stratigraphy, as now known from detailed field and petrochemical correlation, are those given by Watkins and Baksi (1974). Based on these age dates, our preferred correlation of the magnetostratigraphy with the most recent version of the magnetic events of deep-sea sedimentary cores places these early formations of the Columbia River Basalt with N_0 apparently equivalent to magnetic anomaly 5C and N_1 equivalent to magnetic anomaly 5B (fig. 6; LaBrecque, Kent, and Cande, 1977).

Tectonic Rotation

In the Pacific Northwest, major post-Mesozoic tectonic rotations about vertical axes, as inferred by Becker (1932), Cary (1958), and Wise (1963) using diverse arguments, have been supported by the paleomagnetic data of Cox (1957), Watkins (1964), Watkins and Baksi (1974), and Simpson and Cox (1977). Recently Beck (1976) has formulated a modified tectonic model based on "ball-bearing" rotations of blocks

between northwest-southeast lineaments in the Pacific Northwest and has shown that the available paleomagnetic data support the concept. Table 2 and figure 7 present the mean paleomagnetic results for selected groups in this study. The data all feature slight westerly declinations, which, in view of the easterly declinations of most Upper Cenozoic lavas (Wilson, 1971) could be interpreted to represent some small anticlockwise rotation of the region. M. E. Beck (personal commun., 1978) gives the latest estimate of the mid to late Tertiary reference pole as long 115°E, lat 84.5°N with a 95 percent confidence circle of 3.5°. This estimate results in a field direction with inclination = 61° and declination = 354° on the Columbia Plateau. Given the uncertainties reflected in the circles of 95 percent confidence, there is no statistically significant evidence for post-middle Miocene tectonic rotation. The data of Watkins and Baksi (1974) support this conclusion for this particular part of the Columbia Plateau; their conclusions are otherwise for regions immediately to the north, east, and south, although recent work by Choiniere and Swanson (1979) in these regions also finds no evidence of rotation.

CONCLUSIONS

Comparison of laboratory analyses of core samples with the magnetic polarity measured at the outcrop using a fluxgate magnetometer demonstrates the reliability of the portable intrument, if used with appropriate care, and its consequent value in elucidating basalt stratigraphy. Pseudonormal polarity can occur if the samples are taken from the dense interiors of thick flows but can apparently be avoided by sampling the more friable, and oxidized, flow margins.

Detailed mapping using petrochemical data and magnetic polarity measurements have provided a reliable stratigraphic base on which the results of the stable remanent magnetism measurements can be appended. This has resulted in a revision of the earlier magnetostratigraphy set out by Watkins and Baksi (1974) for the Imnaha and Grande Ronde Basalts and more probable correlations with other parts of the Columbia River Basalt sequence.

It is now clear that the Imnaha Basalt flows possess normal polarity (N_0), with the exception of a few basal flows which appear to be re-

TABLE 2

Mean directions of remanent magnetism for each section and polarity magnetozone after demagnetizing treatment

Section or Group	N	D	I	R	α_{95}	θ^1	ϕ^1
GR	9	358.4	63.5	8.5125	13.1	88.5	109.8
RC	15	351.0	63.8	14.5449	7.0	83.7	153.1
DB	6	357.8	55.2	5.7207	16.3	74.8	73.3
All	30	355.6	62.3	28.6970	5.1	86.1	118.7
N_0	11	350.6	58.9	10.5755	9.6	80.7	114.9
R_1	12	354.0	67.1	11.8399	5.3	84.5	199.2
N_1^1	6	356.0	59.7	5.5575	20.8	84.5	210.9

For explanations of sections or groups see figures 3 and 4. For explanation of other symbols see table 1. α_{95} = semivertical angle of 95 percent confidence cone. N = Number of flows.

versed (R_0) and up to four flows at the top which are transitional. The three lowest flows in the overlying sequence of Grande Ronde Basalt are also transitional, and those immediately above are reversed (R_1). The sharp break in magma composition that distinguishes the Imnaha from the Grande Ronde Basalt occurred *during* the polarity transition N_0/R_1. R_1 is followed upward by a normal (N_1) and then another reversed (R_2) sequence of Grande Ronde Basalt flows. Yet another normal (N_2) sequence of Grande Ronde Basalt is found to the west (Swanson and others, 1977) but is correlative in the southeast of the Plateau to a saprolite horizon (Camp, ms).

Taken in conjunction with evidence from other parts of the Columbia Plateau (Swanson and others, in press; Choiniere and Swanson, 1979), there are clearly five and probably seven consecutive polarity epochs represented by the Imnaha, Grande Ronde, and Wanapum Basalts with an age span of approx 16.5 to 13.7 m.y. It is emphasized, however, that the oldest basalt of this sequence, the Imnaha Basalt, has not been reliably dated and that the earlier assumption that it is the

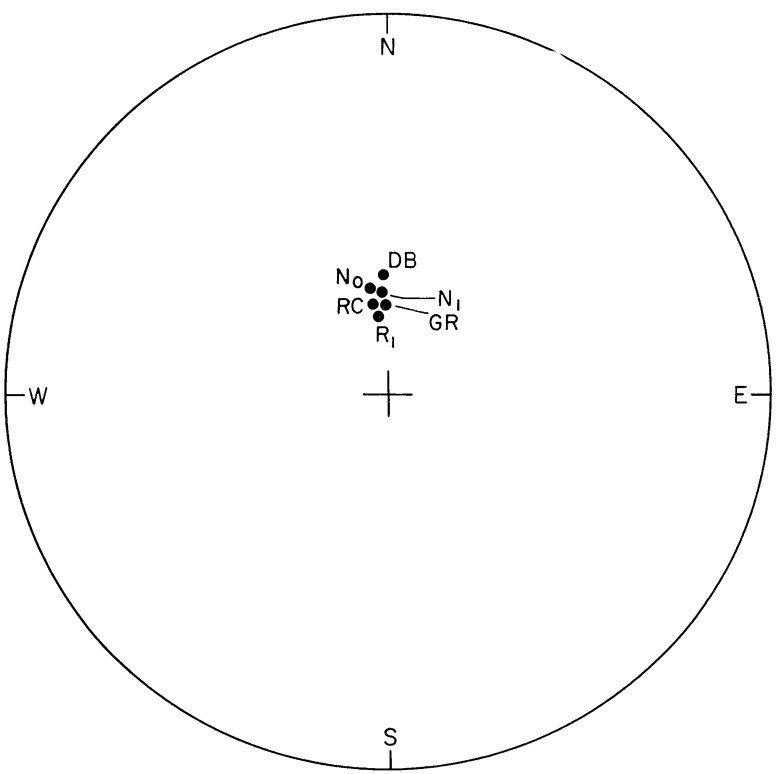


Fig. 7. Mean paleomagnetic directions for selected groups of data: Dug Bar (DB); Grande Ronde (GR); Rocky Canyon (RC); and epochs N_0, R_1, and N_1. The results show no significant difference between any of the sections or epochs. Reversed polarity data are transposed to the northern hemisphere. See table 3 for detailed results.

same age as the Picture Gorge Basalt of north central Oregon is contrary to the available evidence.

The seven consecutive polarity epochs represented by the lower part of the Columbia River Basalt Group in the southeast of the Columbia Plateau are tentatively correlated with magnetic anomalies in polarity epochs 16, 15, and 14 of deep sea sedimentary core analyses, with N_0 equivalent to magnetic anomaly 5C and N_1 equivalent to magnetic anomaly 5B. Finally, within the relatively small part of the plateau discussed in this paper there is no discernible anomaly in the inferred middle Miocene paleomagnetic pole position to support the concept of a post middle Miocene tectonic rotation.

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